Late Fusion with Modality Dropout for MILK10k Lesion Diagnosis: Paired Clinical and Dermoscopic ConvNeXtV2 Encoders, a Three-Seed Ensemble and Cross-Validated Decision Thresholds

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MILK10k 2025 Lesion Diagnosis benchmark, approach description

1. Summary

We describe the system submitted to the MILK10k Lesion Diagnosis benchmark [1]. Each image type has its own ConvNeXtV2-Tiny encoder [2]: the clinical close-up is processed at 224 $\times$ 224 pixels and the dermoscopic image at 384 $\times$ 384 pixels, both after Shades of Gray colour constancy [3]. The embeddings of the two images, the seven MONET concept scores supplied with each image [4] and the patient metadata are combined by a late fusion head that was trained with modality dropout [5], so the same head also works when an input is missing. Encoders and fusion heads were trained in five lesion-level folds with three training seeds, and the 15 resulting models are averaged at test time. Because the benchmark counts a class as positive when its probability is at least 0.5, per-class decision thresholds were tuned on out-of-fold training predictions and folded into the submitted probabilities by a monotone map. No data outside the MILK10k training release were used; the encoders start from public ImageNet-22k weights.

On the 5,240 training lesions, the out-of-fold three-seed ensemble reached a macro F1 (argmax) of 0.557 and a macro one-versus-rest AUC of 0.920. The benchmark metric, estimated by nested cross-validation with thresholds fitted on four folds and applied to the fifth, was 0.564 with all inputs and 0.457 without the dermoscopic image.

2. Data

The MILK10k training release contains 10,480 images of 5,240 lesions, each with one clinical close-up and one dermoscopic image [1]. Of the diagnoses, 95.7% are confirmed by histopathology. Metadata comprise approximate age (missing for 20 lesions), sex, anatomic site (eight categories including unknown) and a skin tone class from 0 to 5. Every image carries seven MONET concept scores: ulceration or crust, hair, vasculature, erythema, pigmentation, gel or immersion fluid, and skin markings. The eleven diagnostic classes are highly imbalanced, from 2,522 lesions of basal cell carcinoma (48.1%) to nine of other malignant proliferations (MAL_OTH, 0.2%).

The benchmark contains 479 lesions with the same two image types and the same metadata and MONET columns. Its skin tone distribution differs slightly from training: 10.9% of benchmark lesions have tone 0 to 2 (11.8% in training), 56.2% tone 3 (60.6%) and 33.0% tone 4 or 5 (27.7%).

3. Preprocessing

Each image is corrected with Shades of Gray colour constancy with Minkowski norm p = 6 [3], which removes global colour casts caused by illumination and camera differences. The corrected image is resized so that its short side is 1.15 times the network input size (258 pixels for clinical images, 442 for dermoscopy), using bicubic interpolation, and stored as a JPEG of quality 95. At training time, random resized crops are taken from this image. Dermoscopic images additionally receive horizontal and vertical flips and rotations of up to 180 degrees, because orientation carries no meaning under the dermatoscope; clinical images receive horizontal flips and rotations of up to 20 degrees. Both receive mild colour jitter (brightness 0.15, contrast 0.15, saturation 0.1, hue 0.02). At inference the stored image is resized to the input size and normalised with ImageNet mean and standard deviation.

The metadata vector has twelve entries: age, a missing-age indicator, skin tone class, sex and a one-hot encoding of the eight sites. Missing ages are replaced by the median age of the training lesions of the same fold. All inputs to the fusion head are standardised with the mean and standard deviation of the training lesions of that fold.

4. Model

4.1 Image encoders

Each image type has its own ConvNeXtV2-Tiny encoder [2], initialised from the public ImageNet-22k checkpoints (`facebook/convnextv2-tiny-22k-224` for clinical images and `facebook/convnextv2-tiny-22k-384` for dermoscopy). The encoder has a 4×4 stride-4 stem, four stages of 3, 3, 9 and 3 blocks with 96, 192, 384 and 768 channels, global average pooling and layer normalisation, which give a 768-dimensional embedding. A linear classifier with dropout 0.3 maps the embedding to the eleven classes during encoder training. Each encoder has 27.9 million parameters.

Encoders are trained with cross-entropy weighted by the square root of inverse class frequency and label smoothing 0.1, using AdamW (weight decay 0.05) with a one-cycle schedule over 12 epochs and mixed precision. The peak learning rate of the last stage is 5×10^{-5}, and each earlier stage receives 0.8 times the rate of the stage after it (layer-wise decay); the classifier receives ten times the base rate. Stochastic depth with rate 0.2 is used, and an exponential moving average of the weights (decay 0.998) is kept. The batch size is 32 for clinical images and 16 with two accumulation steps for dermoscopy. The checkpoint of the averaged weights with the best macro F1 on a held-out 10% of the training lesions is kept.

At inference, the embedding is averaged over test-time views: eight for dermoscopy (four rotations by 90 degrees, each with and without a horizontal flip) and two for clinical images (original and horizontal flip).

4.2 Late fusion head with modality dropout

Figure 1 shows the complete model. The clinical input concatenates the clinical embedding with the seven MONET scores of the clinical image (775 dimensions); the dermoscopic input is built the same way; the metadata input is the twelve-dimensional vector of Section 3. Each input has its own projection (dropout 0.2 for the image inputs, a linear layer to 256 dimensions, GELU and layer normalisation) and its own linear classifier with dropout 0.3. The logits of the inputs that are present are averaged, and a softmax gives the class probabilities. The fusion head has 410,657 parameters.

During training, the availability mask is resampled for every lesion in every batch: with probability 0.5 all three inputs are present, and otherwise one of the seven non-empty subsets of {clinical, dermoscopy, metadata} is drawn uniformly. The head is trained on the stored embeddings with AdamW (learning rate

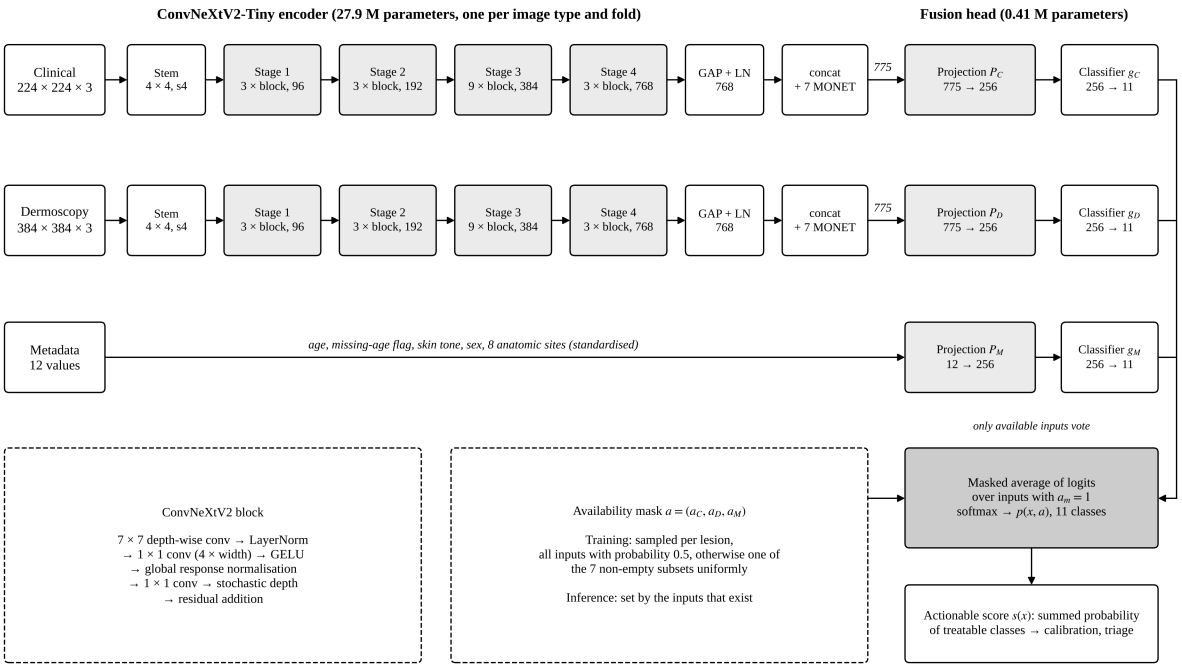


Figure 1: Model. Two ConvNeXtV2-Tiny encoders produce 768-dimensional embeddings that are concatenated with the MONET scores of the same image. Each input is projected to a 256-dimensional token with its own classifier, and the availability mask decides which logits are averaged. During training the mask is sampled by modality dropout.

3×10^{-4}, weight decay 0.05, batch size 256), the same square-root class weights and label smoothing 0.05, for at most 80 epochs. Embeddings of the lesions an encoder was trained on are more separable than those of unseen lesions, so early stopping (patience 15) uses the held-out 10% of lesions that the encoder never trained on, scored by the mean macro F1 over all seven input patterns. For the benchmark submission all three inputs are present.

5. Training protocol

The training release was split into five folds at the lesion level, so that both images of a lesion are always in the same fold, stratified jointly by diagnosis and skin tone group (1,048 lesions per fold). Within each training split, 10% of lesions, stratified by diagnosis, were held out for checkpoint selection and early stopping; this split is fixed by one random seed and identical for every model. For each fold, both encoders and the fusion head were trained with three training seeds (42, 43 and 44), giving 30 encoders and 15 fusion heads in total. Training used one NVIDIA T4 GPU (16 GB) on Kaggle.

6. Inference and decision thresholds

6.1 Ensemble

For each benchmark lesion, the 15 encoder pairs embed its two images with the same preprocessing and test-time views as in training, and the corresponding 15 fusion heads produce class probabilities, which are averaged. As a check of the preprocessing, 24 training lesions were re-embedded from the raw images with the same pipeline; their embeddings matched the stored training embeddings with a cosine similarity of 1.000.

6.2 Thresholds for the benchmark metric

The benchmark scores each class as a binary decision at probability 0.5 and averages the F1 scores (Dice coefficients) over the eleven classes. Well-calibrated multiclass probabilities seldom exceed 0.5 for uncertain lesions, so the plain cut-off under-predicts most classes. For each class k we therefore chose the threshold t_k between 0.02 and 0.98 (step 0.01) that maximises that class's F1 on the out-of-fold training predictions of the three-seed ensemble. Classes with fewer than 20 training lesions (only MAL_OTH) keep $t_k = 0.5$, because a threshold tuned on nine lesions is unreliable. The submitted probability is a monotone piecewise-linear map that sends t_k to 0.5:

$$q_k = \begin{cases} 0.5\, p_k / t_k & p_k < t_k \\ 0.5 + 0.5\,(p_k - t_k)/(1 - t_k) & p_k \geq t_k \end{cases}$$

so that the server's 0.5 rule reproduces the tuned decision while the ranking within each class, and therefore any AUC, is unchanged. Four rules were compared by nested cross-validation, with thresholds fitted on four folds and applied to the fifth (Table 2): the raw 0.5 cut-off, the top class only, tuned thresholds, and tuned thresholds with the top class always positive. Tuned thresholds alone scored best and were used. Table 3 lists the final thresholds, fitted on all 5,240 lesions.

7. Cross-validated results

Table 1 gives the out-of-fold performance on the training release for the input combinations the model supports, for single seeds (mean of three) and for the three-seed ensemble.

Table 1: Out-of-fold performance on the 5,240 training lesions (macro averages over eleven classes; F1 by argmax).

Inputs	Macro F1, single seed	Macro F1, ensemble	Balanced accuracy, ensemble	Macro AUC, single seed	Macro AUC, ensemble
Clinical + dermoscopy + metadata	0.541	0.557	0.547	0.907	0.920
Clinical + dermoscopy	0.518	0.540	0.551	0.892	0.904
Clinical + metadata	0.423	0.441	0.427	0.850	0.872
Dermoscopy only	0.498	0.514	0.536	0.872	0.891
Clinical only	0.399	0.415	0.440	0.808	0.832

Table 2: Benchmark metric (macro F1 with each class positive at 0.5) estimated on the training release by nested cross-validation of the decision rule, three-seed ensemble.

Decision rule	All inputs	Clinical + metadata
Raw probabilities, cut-off 0.5	0.493	0.374
Top class only	0.557	0.441

Decision rule	All inputs	Clinical + metadata
Tuned per-class thresholds (submitted)	0.564	0.457
Tuned thresholds, top class always positive	0.561	0.451

Table 3: Per-class thresholds of the submitted model (all inputs), mapped to 0.5 in the submitted file.

Class	Training lesions	Threshold
AKIEC	303	0.31
BCC	2,522	0.30
BEN_OTH	44	0.19
BKL	544	0.24
DF	52	0.28
INF	50	0.28
MAL_OTH	9	0.50 (not tuned)
MEL	450	0.34
NV	746	0.37
SCCKA	473	0.32
VASC	47	0.31

Averaging three seeds raised macro F1 by 0.016 and macro AUC by 0.013 with all inputs. Dermoscopy is the more informative image: without it, macro F1 falls from 0.557 to 0.441. Threshold tuning adds 0.071 to the benchmark metric compared with the raw 0.5 cut-off. On the benchmark images, the model predicts BCC as the top class for 136 of 479 lesions and SCCKA for 114, which differs from the training prevalence (48.1% and 9.0%) and suggests that the benchmark class distribution differs from that of the training release.

8. Reproducibility

All steps run on a single T4 GPU. Refitting the 15 fusion heads on stored embeddings takes about 3 minutes, and embedding the 479 benchmark lesions with all 30 encoders, including test-time views, takes about 11 minutes. Table 4 summarises the settings.

Table 4: Settings of the submitted system.

Component	Setting
Colour constancy	Shades of Gray, p = 6
Clinical encoder	ConvNeXtV2-Tiny, ImageNet-22k, 224 px, 2 test-time views
Dermoscopy encoder	ConvNeXtV2-Tiny, ImageNet-22k, 384 px, 8 test-time views
Encoder optimisation	AdamW, peak 5×10^{-5}, layer decay 0.8, one-cycle, 12 epochs, EMA 0.998, stochastic depth 0.2, label smoothing 0.1
Class weighting	square root of inverse class frequency

Component	Setting
Fusion head	late fusion, 256-dimensional tokens, modality dropout (0.5 full inputs), 410,657 parameters
Fusion optimisation	AdamW 3×10^{-4}, batch 256, at most 80 epochs, early stopping on held-out lesions
Validation	5 lesion-level folds stratified by diagnosis and skin tone; 10% inner hold-out
Ensemble	3 seeds $\times$ 5 folds = 15 models, probabilities averaged
Decision thresholds	per class, tuned on out-of-fold predictions; 0.5 for classes with fewer than 20 lesions
External data	none (public ImageNet-22k initialisation only)

9. Limitations

The thresholds are tuned for the class distribution of the training release; if the benchmark distribution differs, as the predicted class counts suggest, the optimal thresholds differ too. The rarest classes (MAL_OTH, BEN_OTH, INF, VASC and DF, each with 52 lesions or fewer) contribute equally to the macro metric but are learned from very few examples. MILK10k provides no patient identifier, so lesions of the same patient may fall in different folds and the cross-validated estimates may be slightly optimistic.

References

[1] Tschandl P, Akay BN, Rosendahl C, Rotemberg V, Todorovska V, Weber J, et al. MILK10k: a hierarchical multimodal imaging-learning toolkit for diagnosing pigmented and nonpigmented skin cancer and its simulators. J Invest Dermatol. 2026;146(2):357. doi:10.1016/j.jid.2025.06.1594

[2] Woo S, Debnath S, Hu R, Chen X, Liu Z, Kweon IS, et al. ConvNeXt V2: co-designing and scaling ConvNets with masked autoencoders. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR); 2023. p. 16133–16142.

[3] Finlayson GD, Trezzi E. Shades of gray and colour constancy. In: Proceedings of the 12th Color and Imaging Conference. Society for Imaging Science and Technology; 2004. p. 37–41.

[4] Kim C, Gadgil SU, DeGrave AJ, Omiye JA, Cai ZR, Daneshjou R, et al. Transparent medical image AI via an image–text foundation model grounded in medical literature. Nat Med. 2024;30(4):1154–1165.

[5] Neverova N, Wolf C, Taylor G, Nebout F. ModDrop: adaptive multi-modal gesture recognition. IEEE Trans Pattern Anal Mach Intell. 2016;38(8):1692–1706.